

Problems

5) From the cubics P_3 to the fourth degree polynomials P_4 , what matrix represents multiplication by $3t-5$?

Solution:- Basis for $P_3 = \{v_1=1, v_2=t, v_3=t^2, v_4=t^3\}$

Basis for $P_4 = \{u_1=1, u_2=t, u_3=t^2, u_4=t^3, u_5=t^4\}$

To find Matrix:-

$$(3t-5)v_1 = (3t-5)1 = -5u_1 + 3u_2 + 0 \cdot u_3 + 0 \cdot u_4 + 0 \cdot u_5 \\ = (-5, 3, 0, 0, 0)$$

$$(3t-5)v_2 = (3t-5)t = -5t + 3t^2 = 0 \cdot u_1 - 5u_2 + 3u_3 + 0 \cdot u_4 + 0 \cdot u_5 \\ = (0, -5, 3, 0, 0)$$

$$(3t-5)v_3 = (3t-5)t^2 = -5t^2 + 3t^3 = 0 \cdot u_1 + 0 \cdot u_2 - 5u_3 + 3u_4 + 0 \cdot u_5 \\ = (0, 0, -5, 3, 0)$$

$$(3t-5)v_4 = (3t-5)t^3 = 3t^4 - 5t^3 = 0 \cdot u_1 + 0 \cdot u_2 + 0 \cdot u_3 - 5u_4 + 3u_5 \\ = (0, 0, 0, -5, 3)$$

$$\therefore A_T = \begin{bmatrix} -5 & 0 & 0 & 0 \\ 3 & -5 & 0 & 0 \\ 0 & 3 & -5 & 0 \\ 0 & 0 & 3 & -5 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

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1 ii) Find the image of these points after applying the transformation given :-

i) Reflect $(-3, 2)$ across 90° line and then project on x -axis

ii) Project $(3, 4)$ on y -axis and then rotate by 45° in clockwise direction

Solution i) Reflection matrix to reflect any vector ~~across~~ across the 90° line is

$$H = \begin{bmatrix} 2\cos^2\theta - 1 & 2\cos\theta \cdot \sin\theta \\ 2\cos\theta \cdot \sin\theta & 2\sin^2\theta - 1 \end{bmatrix}$$

$$H_{90^\circ} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Projection matrix to project any vector on x -axis is $P_0 = \begin{bmatrix} \cos^2 0 & \cos 0 \cdot \sin 0 \\ \cos 0 \cdot \sin 0 & \sin^2 0 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

ii) Projection matrix to project any vector on y -axis is $P_{90^\circ} = \begin{bmatrix} \cos^2 90^\circ & \cos 90^\circ \cdot \sin 90^\circ \\ \cos 90^\circ \cdot \sin 90^\circ & \sin^2 90^\circ \end{bmatrix}$

$$P_{90^\circ} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Rotation matrix to rotate any vector by 45° in clockwise direction is :-

$$Q_{-45^\circ} = \begin{bmatrix} \cos(-45^\circ) & -\sin(-45^\circ) \\ \sin(-45^\circ) & \cos(-45^\circ) \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\therefore \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{4}{\sqrt{2}} \\ \frac{4}{\sqrt{2}} \end{bmatrix} = 2\sqrt{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

1 i) Write down the co-ordinates of the image of the point $(3, -2)$ when :-

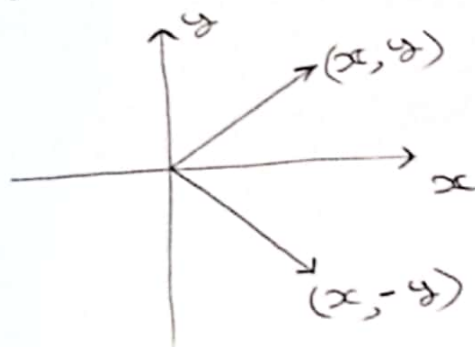
i) reflected in the x-axis

ii) reflected in the y-axis

iii) reflected in the x-axis followed by a reflection in the y-axis

iv) reflected in the origin

Solution :- i) Reflection about x-axis



$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

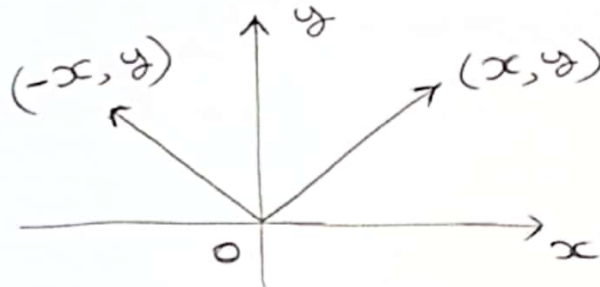
$$\therefore A_T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

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∴ Reflection of $(3, -2)$ about x -axis is

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

i) Reflection about y -axis :-



$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} ; T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} ; \therefore A_T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

∴ Reflection of $(3, -2)$ about y -axis is :-

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$$

iii) Reflection in x -axis followed by reflection in y -axis :-

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

iv) Reflection in the origin :-

4) Find the matrix of the linear transformation ^⑤
 T on $\mathbb{R}^3$ defined by $T(x, y, z) = (x+2y-z, y+z, x+y-2z)$ with respect to $\circ$ -

i) the standard basis $(1, 0, 0)$ $(0, 1, 0)$ $(0, 0, 1)$

ii) the basis $(1, 1, 1)$, $(1, 1, 0)$ $(1, 0, 0)$

Solution i) $T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$; $T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$;

$$T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}$$

$$\therefore A_T = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}$$

ii) $T \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$; $T \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$; $T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$\underbrace{\quad}_{V_1} \qquad \qquad \underbrace{\quad}_{V_2} \qquad \qquad \underbrace{\quad}_{V_3}$

$$V = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = K_1(1, 1, 1) + K_2(1, 1, 0) + K_3(1, 0, 0)$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = (K_1, K_1, K_1) + (K_2, K_2, 0) + (K_3, 0, 0)$$

$$x = K_1 + K_2 + K_3 ; y = K_1 + K_2$$

$$\boxed{z = K_1}$$

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$$\boxed{k_2 = y - z} ; k_1 + k_2 + k_3 = x$$

$$-z + y - z + k_3 = x$$

$$\boxed{k_3 = x - y}$$

$$\therefore V = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = k_1(1, 1, 1) + k_2(1, 1, 0) + k_3(1, 0, 0)$$

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = k_1 \cdot T \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + k_2 T \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + k_3 T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= z(2, 2, 0) + y - z(3, 1, 2) + x - y(1, 0, 1)$$

$$= \begin{pmatrix} 2z \\ 2z \\ 0 \end{pmatrix} + \begin{pmatrix} 3y - 3z \\ y - z \\ 2y - 2z \end{pmatrix} + \begin{pmatrix} x - y \\ 0 \\ x - y \end{pmatrix}$$

$$= \begin{pmatrix} 2z + 3y - 3z + x - y \\ 2z + y - z \\ 2y - 2z + x - y \end{pmatrix}$$

$$= \begin{pmatrix} x + 2y - z \\ y + z \\ x + y - 2z \end{pmatrix}$$

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}}_{A_T} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

2) Find the linear transformation that maps $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $(1, 1)$ to $(-1, 4)$ and $(-1, 3)$ to $(-7, 0)$?

Solution:- $T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$; $T \begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -7 \\ 0 \end{pmatrix}$

$\downarrow$
 V_1

$$V = \begin{pmatrix} x \\ y \end{pmatrix} = K_1 \cdot (1, 1) + K_2 (-1, 3)$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = (K_1, K_1) + (-K_2, 3K_2)$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = (K_1 - K_2, K_1 + 3K_2)$$

$$x = K_1 - K_2$$

$$y = K_1 + 3K_2$$

$$(-) \quad (-) \quad (-)$$

$$x - y = -4K_2$$

$$\boxed{K_2 = \frac{y - x}{4}}$$

$$x = \frac{K_1}{1} - \left\{ \frac{y - x}{4} \right\}$$

$$x = \frac{4K_1 - y + x}{4}$$

$$4x = 4K_1 - y + x$$

$$\boxed{K_1 = \frac{3x + y}{4}}$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \frac{3x + y}{4} (1, 1) + \frac{y - x}{4} (-1, 3)$$

3)

$$\begin{aligned}
 T \begin{pmatrix} x \\ y \end{pmatrix} &= \frac{3x+y}{4} T \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{y-x}{4} T \begin{pmatrix} -1 \\ 3 \end{pmatrix} \\
 &= \frac{3x+y}{4} (-1, 4) + \frac{y-x}{4} (-7, 0) \\
 &= \left(\frac{-3x-y}{4}, \frac{3x+y}{4} \right) + \left(\frac{-7y+7x}{4}, 0 \right) \\
 &= \left(\frac{-3x-y-7y+7x}{4}, \frac{3x+y}{4} \right) \\
 &= \left(\frac{4x-8y}{4}, \frac{3x+y}{4} \right)
 \end{aligned}$$

$$\boxed{T \begin{pmatrix} x \\ y \end{pmatrix} = x-2y, 3x+y}$$

3) For each of the following linear transformations T , find a basis and the dimension of the range and kernel of T :

i) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x+y, x-y, y)$

ii) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (y, 0)$

Solution i) $A \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ x-y \\ y \end{bmatrix} \quad \mathbb{R}^2 \rightarrow \mathbb{R}^3$

To find the basis & dimension of range of T , we need to find the column space:-

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} \textcircled{1} & 1 \\ 0 & \textcircled{-2} \\ 0 & 1 \end{bmatrix}$$

$$C(A) = c_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Range is a 2D plane spanned by vectors $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ & $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ Dimension = 2

To find a kernel of T , we need to find the null space :- (9)

$$\begin{bmatrix} 1 & 1 \\ 0 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$y = 0; -2y = 0$$

$$x + y = 0$$

$x = 0; y = 0$ ∴ kernel is origin in

$\mathbb{R}^2$.

$$ii) T(x, y) = (y, 0) \Rightarrow T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}; T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$C(A) = C_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \text{Range} = x\text{-axis}$$

Null space $y = 0; x = 0$; origin in $\mathbb{R}^2$

6) What multiple of $a = (1, 1, 1)$ is closest to $b = (2, 4, 4)$? Find also the point on the line through b that is closest to a ?

Solution ∴ Let 'p' be the point on the line through $a = (1, 1, 1)$ closest to $b = (2, 4, 4)$

$$\therefore p = a\hat{x} = a \cdot \frac{a^T b}{a^T a} \quad \boxed{\therefore p = \frac{10}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}$$

$$a^T b = [1 \ 1 \ 1] \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} = 10; a^T a = [1 \ 1 \ 1] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3$$

(10)

Let p_1 be the point on the line through b closest to a .

$$p_1 = b \hat{x} = b \cdot \frac{a^T b}{b^T b}$$

$$\therefore b^T b = [2 \ 4 \ 4] \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} = 36$$

$$a^T b = [1 \ 1 \ 1] \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} = 10$$

$$\therefore p_1 = \frac{10}{36} \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 5/9 \\ 1/9 \\ 1/9 \end{bmatrix}$$

9) Find the matrix P that projects every point in $\mathbb{R}^3$ onto the line of intersection of the planes $x - 2y + 3z = 0$ and $y - z = 0$. What are the column space and row space of this matrix?

Solution:- The line of intersection of these planes is:-

$$x - 2y + 3z = 0$$

$$y - z = 0$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix} \longrightarrow$$

$$\boxed{y = z}$$

$$x - 2y + 3z = 0$$

$$x - 2z + 3z = 0$$

$$\boxed{x = -z}$$

$$\circ \circ \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -z \\ z \\ z \end{pmatrix} = z \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

$\circ \circ$ The line passing through $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$ is the point of intersection.

Let ' a ' = $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$ and the projection matrix through ' a ' is $P = \frac{a \cdot a^T}{a^T \cdot a}$

$$a \cdot a^T = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$a^T a = \begin{bmatrix} -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = 3$$

$$\circ \circ \quad P = \frac{1}{3} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

P is a symmetric matrix of rank 1

$\circ \circ$ Column space and row space of this matrix

are $\left[c_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}; c_1 \in \mathbb{R} \right]$; $C(P)$ & $C(P^T)$ is a line in $\mathbb{R}^3$.

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8) Let P be the plane in $\mathbb{R}^4$ with equation $x+y+z+t=0$. What is the basis for $P^\perp$?
What matrix has the plane P as its null space?

Solution:- A vector perpendicular to plane is nothing but vector normal to that plane.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = 0 ; x = \text{pivot variable} \\ y, z, t = \text{free variables}$$

$$x = -y - z - t$$

$$\therefore \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} -y - z - t \\ y \\ z \\ t \end{pmatrix} = y \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\therefore N(A) = y \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$N(A)$ is perpendicular to $C(AT)$

$\therefore$ Basis for $C(AT)$ or Basis for $P^\perp$ is $(1, 1, 1, 1)$

10) Let $A = \begin{bmatrix} 2 & -2 & 1 \end{bmatrix}$ and let V be the nullspace of A .

Find i) a basis for V and a basis for $V^\perp$ ii) a projection matrix P_1 onto $V^\perp$ iii) the projection matrix P_2 onto V .

Solution :- $\begin{bmatrix} 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$

$$2x - 2y + z = 0$$

let $y = K_1$ & $z = K_2$

$$\therefore 2x - 2K_1 + K_2 = 0$$

$$2x = 2K_1 - K_2$$

$$x = K_1 - K_2/2$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} K_1 - K_2/2 \\ K_1 \\ K_2 \end{pmatrix} = K_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + K_2 \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix}$$

Basis for $V = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}; \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix}$

If V is $N(A)$ then $V^\perp$ is $C(AT)$

$\therefore$ Basis for $V^\perp$ is $\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = V_1$

ii) Projection Matrix P_1 onto $V^\perp$ is given by :-

$$P_1 = \frac{V_1 \cdot V_1^T}{V_1^T \cdot V_1} = \frac{\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 \end{bmatrix}}{\begin{bmatrix} 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}} = \frac{\begin{bmatrix} 4 & -4 & 2 \\ -4 & 4 & -2 \\ 2 & -2 & 1 \end{bmatrix}}{4 + 4 + 1} = \frac{1}{9} \begin{bmatrix} 4 & -4 & 2 \\ -4 & 4 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

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iii) the projection matrix P_2 onto V :- $V = N(A)$ has two basis vectors

$$v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}; v_3 = \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix}$$

$$P_2 = \frac{v_2 \cdot v_2^T}{v_2^T \cdot v_2} = \frac{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}}{\begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}} = \frac{\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}{2}$$

$$P_2 = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$P_3 = \frac{v_3 \cdot v_3^T}{v_3^T \cdot v_3} = \frac{\begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} -1/2 & 0 & 1 \end{bmatrix}}{\begin{bmatrix} -1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix}} = \frac{\begin{bmatrix} 1/4 & 0 & -1/2 \\ 0 & 0 & 0 \\ -1/2 & 0 & 1 \end{bmatrix}}{5/4}$$

$$P_3 = \frac{4}{5} \begin{bmatrix} 1/4 & 0 & -1/2 \\ 0 & 0 & 0 \\ -1/2 & 0 & 1 \end{bmatrix}$$

11) If S is the subspace of $\mathbb{R}^3$ containing only the zero vector, what is $S^\perp$? If S is spanned by $(1, 1, 1)$ what is $S^\perp$? If S is spanned by $(2, 0, 0)$ and $(0, 0, 3)$, what is $S^\perp$?

Solution i) $S^\perp$ is $\mathbb{R}^3$

$$\text{ii)} \quad A^T = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$\text{let } z = k_2 ; y = k_1$$

$$x + y + z = 0$$

$$x + k_1 + k_2 = 0$$

$$x = -k_1 - k_2$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -k_1 - k_2 \\ k_1 \\ k_2 \end{pmatrix} = k_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\therefore \text{Basis for } S^\perp \quad \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \text{ \& } \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{iii)} \quad A^T = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

x, z = pivot variables

y = free variable

$$3z = 0 \Rightarrow z = 0$$

$$2x = 0 \Rightarrow x = 0$$

$$y = k$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ k \\ 0 \end{pmatrix} = k \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\therefore \text{Basis for } S^\perp \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

(16)

13) Project $b = (1, 0, 0)$ onto the lines through $a_1 = (-1, 2, 2)$, $a_2 = (2, 2, -1)$ and $a_3 = (2, -1, 2)$. Add the three projections to get the sum b . Also find the corresponding projection matrices P_1, P_2 and P_3 . Check that their sum is I and the product is O .

Solution:- $b = (1, 0, 0)$

$$P_1 = \frac{a_1^T \cdot b}{a_1^T \cdot a_1} \cdot a_1 = \frac{[-1 \ 2 \ 2] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}{[-1 \ 2 \ 2] \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}} \cdot \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$$

$$P_1 = \frac{-1}{9} \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/9 \\ -2/9 \\ -2/9 \end{bmatrix}$$

$$P_2 = \frac{a_2^T \cdot b}{a_2^T \cdot a_2} \cdot a_2 = \frac{[2 \ 2 \ -1] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}{[2 \ 2 \ -1] \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}} \cdot \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} = \frac{2}{9} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$$

$$P_2 = \begin{bmatrix} 4/9 \\ 4/9 \\ -2/9 \end{bmatrix}$$

$$P_3 = \frac{a_3^T \cdot b}{a_3^T \cdot a_3} \cdot a_3 = \frac{[2 \ -1 \ 2] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}{[2 \ -1 \ 2] \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}} \cdot \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = \frac{2}{9} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

$$P_3 = \begin{bmatrix} 4/9 \\ -2/9 \\ 4/9 \end{bmatrix}$$

$$P_1 + P_2 + P_3 = \begin{bmatrix} 1/9 \\ -2/9 \\ -2/9 \end{bmatrix} + \begin{bmatrix} 4/9 \\ 4/9 \\ -2/9 \end{bmatrix} + \begin{bmatrix} 4/9 \\ -2/9 \\ 4/9 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

we should find the corresponding projection matrices :-

$$\text{Projection Matrix } P_1 = \frac{a_1 \cdot a_1^T}{a_1^T \cdot a_1}$$

$$P_1 = [-1 \ 2 \ 2] \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} = a_1^T \cdot a_1 = 9$$

$$P_1 = \frac{1}{9} \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} [-1 \ 2 \ 2] = \frac{1}{9} \begin{bmatrix} 1 & -2 & -2 \\ -2 & 4 & 4 \\ -2 & 4 & 4 \end{bmatrix}$$

$$\text{Projection matrix } P_2 = \frac{a_2 \cdot a_2^T}{a_2^T \cdot a_2}$$

$$a_2^T \cdot a_2 = [2 \ 2 \ -1] \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} = 9$$

$$a_2 \cdot a_2^T = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} [2 \ 2 \ -1] = \begin{bmatrix} 4 & 4 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 1 \end{bmatrix}$$

$$\therefore P_2 = \frac{1}{9} \begin{bmatrix} 4 & 4 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 1 \end{bmatrix}$$

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Projection Matrix $P_3 = \frac{a_3 \cdot a_3^T}{a_3^T \cdot a_3}$

$$a_3^T \cdot a_3 = [2 \ -1 \ 2] \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = 9$$

$$a_3 \cdot a_3^T = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} [2 \ -1 \ 2] = \begin{bmatrix} 4 & -2 & 4 \\ -2 & 1 & -2 \\ 4 & -2 & 4 \end{bmatrix}$$

$$\therefore P_3 = \frac{1}{9} \begin{bmatrix} 4 & -2 & 4 \\ -2 & 1 & -2 \\ 4 & -2 & 4 \end{bmatrix}$$

Let us consider $P_1 + P_2 + P_3$

$$P_1 + P_2 + P_3 = \begin{bmatrix} 1/9 & -2/9 & -2/9 \\ -2/9 & 4/9 & 4/9 \\ -2/9 & 4/9 & 4/9 \end{bmatrix} + \begin{bmatrix} 4/9 & 4/9 & -2/9 \\ 4/9 & 4/9 & -2/9 \\ -2/9 & -2/9 & 1/9 \end{bmatrix}$$

$$+ \begin{bmatrix} 4/9 & -2/9 & 4/9 \\ -2/9 & 1/9 & -2/9 \\ 4/9 & -2/9 & 4/9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore P_1 + P_2 + P_3 = I$$

we can verify that $P_1 \cdot P_2 \cdot P_3 = 0$

7) Find the projection of $b = (1, 2, 7)$ onto the column space of A spanned by $(1, 1, -2)$ and $(1, -1, 4)$.
 Split $b = p + q$ with p in $C(A)$ and q in $N(A^T)$.

Solution :- Let ' p ' be the projection of ' b ' onto column space of A spanned by vectors $(1, 1, -2)$ and $(1, -1, 4)$.

$$p = A\hat{x}$$

$$\hat{x} = (A^T A)^{-1} \cdot A^T b$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 1 & -2 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 6 & -8 \\ -8 & 18 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & -2 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix} = \begin{bmatrix} -11 \\ 27 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 6 & -8 \\ -8 & 18 \end{bmatrix}^{-1} \begin{bmatrix} -11 \\ 27 \end{bmatrix} = \frac{1}{44} \begin{bmatrix} 18 & 8 \\ 8 & 6 \end{bmatrix} \begin{bmatrix} -11 \\ 27 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 9/22 \\ 37/22 \end{bmatrix}$$

$$p = A \cdot \hat{x} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 9/22 \\ 37/22 \end{bmatrix} = \begin{bmatrix} 23/11 \\ -14/11 \\ 65/11 \end{bmatrix}$$

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$$b = p + q$$

$$q = b - p = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix} - \begin{bmatrix} 23/11 \\ -14/11 \\ 65/11 \end{bmatrix} = \begin{bmatrix} 1 - 23/11 \\ 2 + 14/11 \\ 7 - 65/11 \end{bmatrix}$$

$$q = \begin{bmatrix} -12/11 \\ 36/11 \\ 12/11 \end{bmatrix} ; q \in N(CAT)$$

15) An ice-cream vendor records the number of hours of sun shine (x) versus the number of ice-creams sold in a hour (y) at his shop from Monday to Friday and found the following data:

x :	2	3	5	7	9
y :	4	5	7	10	15

Find the best values of m & c that suit the equation $y = mx + c$. If there is a weather forecast that says there would be 8 hours of sun shine the next day, estimate the number of ice-creams that he expects to sell on that day.

Solution :- $y = mx + c$

$$x = 2, 3, 5, 7, 9 ; y = 4, 5, 7, 10, 15$$

$$y = m(2) + c = 4 \Rightarrow 2m + c = 4$$

$$3(m) + c = 5 \Rightarrow 3m + c = 5$$

$$m(5) + c = 7 \Rightarrow 5m + c = 7$$

$$m(7) + c = 10 \Rightarrow 7m + c = 10$$

$$m(9) + c = 15 \Rightarrow 9m + c = 15$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 7 \\ 1 & 9 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \\ 10 \\ 15 \end{bmatrix}$$

to find the least squares solution $\hat{x}$:-

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$$(A^T A) \hat{x} = A^T b$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 7 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 7 \\ 1 & 9 \end{bmatrix} = \begin{bmatrix} 5 & 26 \\ 26 & 168 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 7 & 9 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 7 \\ 10 \\ 15 \end{bmatrix} = \begin{bmatrix} 41 \\ 263 \end{bmatrix}$$

$$\hat{x} = (A^T A)^{-1} A^T b = \frac{1}{164} \begin{bmatrix} 168 & -26 \\ -26 & 5 \end{bmatrix} \begin{bmatrix} 41 \\ 263 \end{bmatrix}$$

$$= \frac{1}{164} \begin{bmatrix} 50 \\ 249 \end{bmatrix} = \begin{bmatrix} 0.305 \\ 1.518 \end{bmatrix}$$

$$\boxed{y = 1.518x + 0.305}$$

When there would be 8 hours of sunshine, the number of ice creams he expects to sell is :-

$$y = 1.518(8) + 0.305$$

$$\boxed{y = 13}$$

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14) Find $\|E\|^2 = \|Ax - b\|^2$ and solve the normal equations $A^T A \cdot \hat{x} = A^T b$. Find the solution $\hat{x}$ and the projection $p = A \cdot \hat{x}$. (Use least squares method)

Given $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \\ -1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix}$

Solution :-

$$A^T A \cdot \hat{x} = A^T b$$

$$\hat{x} = (A^T A)^{-1} \cdot A^T b$$

$$A^T A = \begin{bmatrix} 2 & 1 & 0 & -1 \\ 1 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 1 & 3 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 2 & 1 & 0 & -1 \\ 1 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 6 & 1 \\ 1 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 8 \\ 0 \end{bmatrix} = \frac{1}{17} \begin{bmatrix} 3 & -1 \\ -1 & 6 \end{bmatrix} \begin{bmatrix} 8 \\ 0 \end{bmatrix} = \begin{bmatrix} 24/17 \\ -8/17 \end{bmatrix}$$

$$p = A \cdot \hat{x} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 24/17 \\ -8/17 \end{bmatrix} = \begin{bmatrix} 40/17 \\ 24/17 \\ 8/17 \\ -32/17 \end{bmatrix} = \frac{8}{17} \begin{bmatrix} 5 \\ 3 \\ 1 \\ -4 \end{bmatrix}$$

12) Find the point on the plane $2x+5y+z=0$ (23) that is nearest to $(2,0,1)$?

Solution $2x+5y+z=0$

$$2x = -5y - z$$

$$x = -\frac{5}{2}y - \frac{z}{2}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{5}{2}y - \frac{z}{2} \\ y \\ z \end{pmatrix} = y \begin{pmatrix} -5/2 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix}$$

$\therefore$ The two basis vectors are $\begin{pmatrix} -5/2 \\ 1 \\ 0 \end{pmatrix}$ & $\begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix}$

$$\text{Let } y = z = 2$$

$$\therefore \text{Basis } \begin{pmatrix} -5 \\ 2 \\ 0 \end{pmatrix} \text{ \& } \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} -5 & -1 \\ 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\hat{x} = (A^T A)^{-1} A^T b; A^T A = \begin{pmatrix} -5 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -5 & -1 \\ 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{bmatrix} 29 & 5 \\ 5 & 5 \end{bmatrix}$$

$$A^T b = \begin{pmatrix} -5 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -10 \\ 0 \end{pmatrix}$$

$$\begin{aligned} \therefore \hat{x} &= \begin{bmatrix} 29 & 5 \\ 5 & 5 \end{bmatrix}^{-1} \begin{pmatrix} -10 \\ 0 \end{pmatrix} = \frac{1}{120} \begin{bmatrix} 5 & -5 \\ -5 & 29 \end{bmatrix} \begin{pmatrix} -10 \\ 0 \end{pmatrix} \\ &= \frac{1}{120} \begin{bmatrix} -50 \\ 50 \end{bmatrix} = \begin{bmatrix} -5/12 \\ 5/12 \end{bmatrix} \end{aligned}$$

$$\therefore p = A \cdot \hat{x} = \begin{bmatrix} -5 & -1 \\ 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -5/12 \\ 5/12 \end{bmatrix} = \begin{bmatrix} 20/12 \\ -10/12 \\ 10/12 \end{bmatrix}$$

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Assignment - unit 3 - Set 01

4) A middle-aged man is stretched on a rack with various forces. Given the measurements of length $L = 5, 6, 7$ (in ft) at forces $F = 1, 2, 4$ (in tons) and assuming that Hooke's law $a + bF$ applies, use least squares to find the man's length 'a' when no force is applied.

Solution :- $L = a + bF$

$$\text{for } F = 1 ; L = a + b = 5$$

$$\text{for } F = 2 ; L = a + b(2) = 6 \Rightarrow a + 2b = 6$$

$$\text{for } F = 4 ; L = a + b(4) = 7 \Rightarrow a + 4b = 7$$

$$\therefore \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}$$

least squares solution $A^T \cdot A \cdot \hat{x} = A^T b$

$$\hat{x} = (A^T \cdot A)^{-1} \cdot A^T b$$

$$A^T \cdot A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 7 & 21 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix} = \begin{bmatrix} 18 \\ 45 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 3 & 7 \\ 7 & 21 \end{bmatrix}^{-1} \begin{bmatrix} 18 \\ 45 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 21 & -7 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 18 \\ 45 \end{bmatrix}$$

$$\hat{x} = \frac{1}{14} \begin{bmatrix} 21 & -7 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 18 \\ 45 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 63 \\ 9 \end{bmatrix} = \begin{bmatrix} 63/14 \\ 9/14 \end{bmatrix} \quad (25)$$

$${}^{\circ} L = \frac{63}{14} + \frac{9}{14} \cdot F$$

when no force is applied $\Rightarrow F=0$

$${}^{\circ} L = 4.5 \text{ feet}$$

3) If S is the subspace of $\mathbb{R}^3$ containing only the zero vector, what is $S^\perp$? If S is spanned by $(1,1,1)$, what is $S^\perp$? If S is spanned by $(2,0,0)$ and $(0,0,3)$, what is $S^\perp$?

Solution i) $S^\perp$ is $\mathbb{R}^3$

$${}^{\circ} i) \quad A^T = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$x + y + z = 0 ; \text{ let } y = K_1 \text{ \& } z = K_2$$

$$x + K_1 + K_2 = 0$$

$$x = -K_1 - K_2$$

$${}^{\circ} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -K_1 - K_2 \\ K_1 \\ K_2 \end{pmatrix} = K_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + K_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$${}^{\circ} \text{Basis for } S^\perp = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \text{ \& } \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$${}^{\circ} ii) \quad A^T = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x, z = \text{pivot variables}; y = \text{free variable}$

$$x=0, y=0+K, z=0$$

$${}^{\circ} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ K \\ 0 \end{pmatrix} = K \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad {}^{\circ} \text{Basis for } S^\perp = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

1) If T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^4$ such that $T(1, 3, 1) = (4, 4, 0, 2)$ and $T(4, 0, 5) = (4, 5, -1, 5)$ find $T(-7, 3, -9)$?

Solution :- $(x, y, z) = k_1(1, 3, 1) + k_2(4, 0, 5)$

$$(x, y, z) = (k_1, 3k_1, k_1) + (4k_2, 0k_2, 5k_2)$$

$$(x, y, z) = (k_1 + 4k_2, 3k_1, k_1 + 5k_2)$$

$$x = k_1 + 4k_2; y = 3k_1; z = k_1 + 5k_2$$

$$\begin{array}{rcl} k_1 + 4k_2 & = & x \\ k_1 + 5k_2 & = & z \\ (-) & (-) & (-) \\ \hline -k_2 & = & x - z \\ \hline k_2 & = & z - x \end{array}$$

$$\begin{array}{rcl} k_1 + 4\{z - x\} & = & x \\ k_1 + 4z - 4x & = & x \\ k_1 & = & x + 4x - 4z \\ \hline k_1 & = & 5x - 4z \end{array}$$

$$(x, y, z) = 5x - 4z(1, 3, 1) + (z - x)(4, 0, 5)$$

$$\frac{16 \times 7}{112}$$

$$T(x, y, z) = 5x - 4z T(1, 3, 1) + (z - x) T(4, 0, 5)$$

$$T(x, y, z) = 5x - 4z(4, 4, 0, 2) + (z - x)(4, 5, -1, 5)$$

$$T(x, y, z) = (20x - 16z, 20x - 16z, 0, 10x - 8z) + (4z - 4x, 5z - 5x, x - z, 5z - 5x)$$

$$T(x, y, z) = (16x - 12z, 15x - 11z, x - z, 5x - 3z)$$

$$\therefore T(-7, 3, -9) = 16(-7) - 12(-9), 15(-7) - 11(-9), -7 + 9, 5(-7) - 3(-9)$$

$$\Rightarrow T(-7, 3, -9) = -112 + 108, -105 + 99, 2, -35 + 27$$

$$\Rightarrow T(-7, 3, -9) = -4, -6, 2, -8$$

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2) Suppose T is reflection across the 45 degree line and S is reflection across the y -axis. If $v = (2, 1)$ and $T(v) = (1, 2)$. Find $S(T(v))$ and $T(S(v))$. Show that ST not equal to TS .

Solution $T = H_{45} = \begin{bmatrix} 2\cos^2(45) - 1 & 2\cos(45)\sin(45) \\ 2\cos(45)\sin(45) & 2\sin^2(45) - 1 \end{bmatrix}$

$$= \begin{bmatrix} 2\left(\frac{1}{2}\right) - 1 & 2\left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) \\ 2\left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) & 2\left(\frac{1}{2}\right) - 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$T(v) = (1, 2)$$

$$\circ \circ \quad S(T(v)) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

For reflection across the y -axis, $A_T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\circ \circ \quad S(T(v)) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \Rightarrow S(T(v)) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$S(v) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$T(S(v)) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \Rightarrow T(S(v)) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\circ \circ \quad \boxed{ST \neq TS}$$

Example $T(x, y, z) = (2x + y, 2y - 3z, x - z)$ then the pre image of $\omega = (4, 1, -1)$ is

Solution

$$2x + y = 4$$

$$2y - 3z = 1$$

$$x - z = -1$$

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & -3 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix}$$

$$[A|b] = \left[\begin{array}{ccc|c} 2 & 1 & 0 & 4 \\ 0 & 2 & -3 & 1 \\ 1 & 0 & -1 & -1 \end{array} \right] \xrightarrow{R_3 - \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 2 & 1 & 0 & 4 \\ 0 & 2 & -3 & 1 \\ 0 & -\frac{1}{2} & -1 & -3 \end{array} \right]$$

$$(-\frac{1}{2}) - x(2) = 0$$

$$-\frac{1}{2} = 2x$$

$$\boxed{x = -\frac{1}{4}}$$

$$\left[\begin{array}{ccc|c} 2 & 1 & 0 & 4 \\ 0 & 2 & -3 & 1 \\ 0 & -\frac{1}{2} & -1 & -3 \end{array} \right] \xrightarrow{R_3 + \frac{1}{4}R_2} \left[\begin{array}{ccc|c} 2 & 1 & 0 & 4 \\ 0 & 2 & -3 & 1 \\ 0 & 0 & -\frac{7}{4} & -\frac{11}{4} \end{array} \right]$$

$$\frac{-\frac{1}{4} - \frac{3}{4}}{-4 - 3} = \frac{-1}{-7}$$

$$\frac{-\frac{3}{4} + \frac{1}{4}}{-\frac{12}{4} + \frac{1}{4}} = \frac{-\frac{2}{4}}{-\frac{11}{4}}$$

$$2x + y = 4$$

$$2y - 3z = 1$$

$$\frac{-7}{4}z = -\frac{11}{4}$$

$$\boxed{z = \frac{11}{7}}$$

$$2y - 3\left\{\frac{11}{7}\right\} = 1$$

$$2y - \frac{33}{7} = 1$$

$$2y = \frac{1}{1} + \frac{33}{7}$$

$$2y = \frac{7 + 33}{7}$$

$$2y = \frac{40}{7} ; \boxed{y = \frac{20}{7}}$$

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$$2x + y = 4$$

$$2x + \frac{20}{7} = 4$$

$$2x = \frac{4}{1} - \frac{20}{7}$$

$$2x = \frac{28-20}{7}$$

$$2x = \frac{8}{7}$$

$$\boxed{x = \frac{4}{7}}$$

$$\therefore x = \frac{4}{7} ; y = \frac{20}{7} \text{ or } z = \frac{11}{7}$$